The Algorithmic Monoculture Paradox: Balancing Individual Benefit and Collective Error in Foundation-Model-Mediated Networks

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Abstract

The "Wisdom of Crowds" principle establishes that distributed networks can achieve high collective accuracy through idiosyncratic error cancellation. Today, the rapid integration of foundation models into decision workflows fundamentally alters the epistemic structure of networked deliberation. While incorporating foundation-model advice produces substantial reductions in individual expected squared error—reducing individual error by over 12-fold in task-persistent settings (from 225.0 to 18.37) and up to 46-fold under dynamic sampling (to 4.89)—correlated machine errors ($\rho = 0.90$) accumulate in the collective consensus. Combining the Krogh-Vedelsby / Page decomposition with iterated DeGroot dynamics validated across 199,000 Monte Carlo runs ($N = 1,000$ agents, $T = 30$ rounds), we analyze this tension. In idealized orthogonal crowds ($\rho_h = 0$), monoculture causes collective error to escalate 57-fold (from 0.327 ± 0.009 to 18.648 ± 0.468), though crowd error remains $12\times$ lower than unassisted individual error. Crucially, because initial reliance reduces error ($\frac{\partial E}{\partial \alpha}\big|_{\alpha=0} = -2T \cdot \mathrm{MSE}_{\mathrm{human}} < 0$), an interior optimum exists even for orthogonal humans; when humans exhibit correlated cognitive heuristics ($\rho_h > 0$), this negative gradient is amplified, establishing a pronounced U-shaped trade-off. Socially optimal per-step reliance ($\alpha^* \in [0.008, 0.056]$, cumulative $w_{\mathrm{cum}} \in [20\%, 82\%]$) reduces collective error by 20%–83%. When decentralized agents uncooperatively adopt isolated benchmarks, the stylized Epistemic Price of Anarchy / Overshoot Ratio spans PoA $\in [1.2, 4.1]$, reaching up to $8.0\times$ at complete per-step reliance ($\alpha_{\mathrm{step}} = 1.0$). When $\rho_h \geq 8.08\%$, even pure AI ($\alpha_{\mathrm{step}} = 1.0$) outperforms unassisted humans, rendering excessive reliance an epistemic overshoot rather than a degradation. We prove analytically that uniform external advice passes invariant through row-stochastic networks ($\mathbf{W1} = \mathbf{1}$) under homogeneous reliance, and show that multi-model competition ($\rho = 0.50$) provides limited protection (error ≈ 11.6 vs. 18.6 under monopoly monoculture).

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1 Introduction

For over a century, mathematical statistics and social epistemology have recognized that collective intelligence emerges from cognitive diversity (Galton, 1907; Krogh and Vedelsby, 1995; Page, 2007; Lorenz et al., 2011; Becker, Brackbill, and Centola, 2017). When individuals deliberate across communication topologies, idiosyncratic errors cancel out under linear aggregation, allowing the crowd's collective judgment to outperform even its most capable members.

The rapid worldwide deployment of generative foundation models (Large Language Models, or LLMs) has disrupted this epistemic architecture (Bommasani et al., 2022). Today, millions of professionals, researchers, students, and autonomous software agents consult a small cluster of centralized frontier models for decision support, text synthesis, and heuristic guidance.

Under the parameterized error distributions considered here, incorporating foundation-model advice produces substantial reductions in individual expected squared error: querying an aligned foundation model resolves ambiguities and provides strong local variance reduction. However, evaluating AI assistance exclusively through individual benchmarks commits an ecological fallacy. Building on the theory of algorithmic monoculture formalized by Kleinberg and Raghavan (2021), when independent decision-makers adopt identical or highly correlated algorithmic rules, the system exhibits Algorithmic Monoculture.

Recent empirical audits confirm that commercial frontier foundation models exhibit substantial shared failure modes across diverse architectures: for instance, Kim et al. (2025) evaluate over 350 LLMs on a popular leaderboard benchmark and observe that distinct models agree on identical incorrect multiple-choice answers approximately 60% of the time conditional on both failing. While discrete classification overlap differs from continuous scalar error correlation, it highlights that algorithmic outputs fail to act as independent signals. Meanwhile, Wu, Black, and Chandrasekaran (2025) sug-

gest that post-training alignment processes systematically homogenize model output distributions. Concurrently, behavioral studies demonstrate that generative assistance can reduce the collective diversity of novel content across human creators (Doshi and Hauser, 2024). In real-world environments, humans do not interact with algorithms in isolation; they deliberate, exchange beliefs, and form consensus across structured communication networks (Watts and Strogatz, 1998; Barabási and Albert, 1999; Golub and Jackson, 2010; Zollman, 2007). In human crowds, social influence, shared heuristics, and common media sources induce baseline error correlation $\rho_h > 0$ (Lorenz et al., 2011; Soll and Larrick, 2009; Becker, Brackbill, and Centola, 2017); in our analysis, we evaluate illustrative stylized regimes of modest to moderate correlated human error ($\rho_h \in [0.02, 0.10]$).

This paper addresses three central open questions:

1. How does algorithmic reliance (α) interact with human cognitive correlation (ρ_h) and inter-model correlation (ρ) in networked crowds?

2. Under what conditions does foundation model advice improve versus degrade collective accuracy?

3. Why does an isolated decision-maker's private reliance benchmark overshoot the socially optimal reliance level in deliberating networks?

Through a unified analytical framework verified by 199,000 high-throughput Monte Carlo simulation runs across $1,000$-agent networks ($K = 1,000$ independent trials per condition), we combine the Krogh-Vedelsby / Page ensemble error decomposition (Krogh and Vedelsby, 1995; Page, 2007) with iterated DeGroot consensus dynamics under persistent external influence (DeGroot, 1974; Friedkin and Johnsen, 1990; Acemoglu et al., 2013). Our findings demonstrate that algorithmic monoculture creates an epistemic dilemma: while moderate AI reliance anchors human crowds subject to correlated estimation errors, private optimization benchmarks encourage reliance levels that drive the network into a regime of collective epistemic overshoot.

2 Theoretical Framework

2.1 Consensus Dynamics and Ambiguity Decomposition

Consider an objective state of the world $\theta \in \mathbb{R}$ estimated by a network of N deliberating agents. Let $\mathbf{x}(t) = [x_1(t), \ldots, x_N(t)]^\top \in \mathbb{R}^N$ denote the vector of agent beliefs at timestep t. Agent beliefs evolve through iterative local communication combined with repeated consultation of external foundation model advice:

$$\mathbf{x}(t+1) = (1-\alpha)\mathbf{W}\mathbf{x}(t) + \alpha\hat{\boldsymbol{\theta}}_M(t) \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{N\times N}$ is an aperiodic, irreducible row-stochastic trust matrix ($\mathbf{W1} = \mathbf{1}$), $\alpha \in [0,1]$ is the per-step algorithmic reliance weight, and $\hat{\boldsymbol{\theta}}_M(t)$ is the external advice vector provided to the agents at round t.

To formalize the interplay between individual error and collective judgment, we employ the Krogh-Vedelsby / Page ambiguity decomposition (Krogh and Vedelsby, 1995; Page, 2007). Let $c(t) = \frac{1}{N}\sum_{i=1}^N x_i(t) = \frac{1}{N}\mathbf{1}^\top\mathbf{x}(t)$ denote the unweighted crowd consensus mean. The collective squared error decomposes identically as:

$$\mathrm{MSE}_{\mathrm{coll}}(t) \equiv (c(t)-\theta)^2 = \overline{\mathrm{MSE}}_{\mathrm{ind}}(t) - \mathrm{Diversity}(t) \tag{2}$$

where $\overline{\mathrm{MSE}}_{\mathrm{ind}}(t) = \frac{1}{N}\sum_{i=1}^N (x_i(t)-\theta)^2$ is the average individual squared error and $\mathrm{Diversity}(t) = \frac{1}{N}\sum_{i=1}^N (x_i(t)-c(t))^2$ is the empirical belief variance of the crowd.

Following parameterized correlated error formulations (Kleinberg and Raghavan, 2021), external advice to agent i at round t is modeled as:

$$\hat{\theta}_{M,i}(t) = \theta + \mu_{AI} + \sqrt{\rho}\,\eta_{\mathrm{shared}}(t) + \sqrt{1-\rho}\,\eta_{\mathrm{indiv},i}(t) \tag{3}$$

where μ_{AI} is systematic directional alignment bias, $\eta_{\mathrm{shared}}(t) \sim \mathcal{N}(0, \sigma_{AI}^2)$ is correlated systemic error shared across querying agents, $\eta_{\mathrm{indiv},i}(t) \sim \mathcal{N}(0, \sigma_{AI}^2)$ is independent idiosyncratic model noise across agents, and $\rho \in [0,1]$ represents the inter-agent error correlation parameter.

2.2 Finite-Time Consensus Dynamics and Topology Invariance

In networked deliberation, collective consensus is measured by the unweighted crowd mean $c(t) = \frac{1}{N}\sum_{i=1}^N x_i(t) = \frac{1}{N}\mathbf{1}^\top\mathbf{x}(t)$. To characterize $c(t)$ in finite time across arbitrary network topologies, we define the finite-time projection vector:

$$\mathbf{q}_k^\top \equiv \frac{1}{N}\mathbf{1}^\top\mathbf{W}^k \quad (k \geq 0) \tag{4}$$

with initial condition $\mathbf{q}_0^\top = \frac{1}{N}\mathbf{1}^\top$. Because $\mathbf{W}$ is row-stochastic ($\mathbf{W1} = \mathbf{1}$), every matrix power satisfies $\mathbf{W}^k\mathbf{1} = \mathbf{1}$. Consequently, the projection vector preserves total probability mass identically at every round:

$$\mathbf{q}_k^\top\mathbf{1} = \frac{1}{N}\mathbf{1}^\top\mathbf{W}^k\mathbf{1} = \frac{1}{N}\mathbf{1}^\top\mathbf{1} = 1 \quad \forall k \geq 0 \tag{5}$$

Unrolling the belief evolution Equation 1 from $t = 0$ to horizon T yields:

$$\mathbf{x}(T) = (1-\alpha)^T\mathbf{W}^T\mathbf{x}(0) + \sum_{k=0}^{T-1}\alpha(1-\alpha)^k\mathbf{W}^k\hat{\boldsymbol{\theta}}_M(T-1-k) \tag{6}$$

Left-multiplying by $\frac{1}{N}\mathbf{1}^\top$ produces the exact finite-time arithmetic consensus:

$$c(T) = (1-\alpha)^T \mathbf{q}_T^\top \mathbf{x}(0) + \sum_{k=0}^{T-1} \alpha(1-\alpha)^k \mathbf{q}_k^\top \hat{\boldsymbol{\theta}}_M(T-1-k) \tag{7}$$

Decomposing external advice into uniform shared advice and individual residual noise:

$$\hat{\boldsymbol{\theta}}_M(t) = \theta_0(t)\mathbf{1} + \sqrt{1-\rho}\,\boldsymbol{\eta}_{\text{indiv}}(t) \tag{8}$$

where $\theta_0(t) = \theta + \mu_{AI} + \sqrt{\rho}\,\eta_{\text{shared}}(t)$. By the stochastic identity (Equation 5), $\mathbf{q}_k^\top(\theta_0(t)\mathbf{1}) = \theta_0(t)(\mathbf{q}_k^\top\mathbf{1}) = \theta_0(t)$ for every round k. Consequently, the uniform shared advice mode decouples identically from the network topology $\mathbf{W}$ at every finite horizon T. The cumulative weight on external advice over T deliberation rounds defines cumulative reliance:

$$\boxed{w_{\text{cum}} = 1 - (1-\alpha)^T \equiv b_T} \tag{9}$$

Because $\sum_{k=0}^{T-1}\alpha(1-\alpha)^k = b_T$, the expected consensus bias is strictly topology-invariant: $\mathbb{E}[c(T) - \theta] = \mu_{AI} b_T$.

Meanwhile, let $\mathbf{P} = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ denote the centering projector such that $\text{Diversity}(t) = \frac{1}{N}\|\mathbf{P}\mathbf{x}(t)\|_2^2$. Because $\mathbf{W}^k\mathbf{1} = \mathbf{1}$ and $\mathbf{P}\mathbf{1} = \mathbf{0}$, the uniform shared advice mode vanishes ($\mathbf{P}\mathbf{W}^k\mathbf{1} = \mathbf{0}$). Under dynamic sampling where $\boldsymbol{\eta}_{\text{indiv}}(t)$ is independent across rounds, asymptotic belief diversity is given exactly by:

$$\lim_{t\to\infty}\mathbb{E}[\text{Diversity}(t)] = \frac{\alpha^2(1-\rho)\sigma_{AI}^2}{N}\sum_{k=0}^{\infty}(1-\alpha)^{2k}\text{Tr}\big(\mathbf{P}\mathbf{W}^k(\mathbf{W}^\top)^k\mathbf{P}\big) \tag{10}$$

At $\alpha = 1.0$, only the $k = 0$ term survives and steady-state diversity equals $(1-\rho)\sigma_{AI}^2\frac{N-1}{N} \approx (1-\rho)\sigma_{AI}^2$. For $\alpha \in (0, 1)$, iterative social averaging dampens individual noise, scaling diversity strictly below $(1-\rho)\sigma_{AI}^2$ in proportion to α^2.

2.3 Temporal Persistence: Dynamic vs. Static AI Noise

In multi-round deliberation, the temporal accumulation of algorithmic error depends on whether model errors are stochastic across queries or fixed for a given task:

1. Dynamic AI Noise (Temporal Stochasticity): In this regime, agents refresh queries at each round via stochastic temperature draws or track an ongoing information stream where α acts as an exponential moving average (EMA) filter. Here, the shared error $\eta_{\text{shared}}(t)$ is i.i.d. across rounds, and idiosyncratic AI errors are independent across both agents and successive rounds. Taking the expectation of $(c(T)-\theta)^2$ from Equation 7 yields the exact finite-time collective squared error:

$$E_{\text{dynamic}} = (1-\alpha)^{2T}\text{MSE}_{\text{human}}(\rho_h, T) + E_{\text{AI}}^{\text{dyn}}(\alpha, \rho, T) \tag{11}$$

where the exact AI-induced error contribution evaluates to:

$$E_{\text{AI}}^{\text{dyn}} = \mu_{AI}^2 b_T^2 + \frac{\alpha(1-(1-\alpha)^{2T})}{2-\alpha}\rho\sigma_{AI}^2 + \alpha^2(1-\rho)\sigma_{AI}^2\sum_{k=0}^{T-1}(1-\alpha)^{2k}\|\mathbf{q}_k\|^2 \tag{12}$$

and $\text{MSE}_{\text{human}}(\rho_h, T) = \sigma_h^2[\rho_h + (1-\rho_h)\|\mathbf{q}_T\|^2]$ with $\sigma_h^2 = 225.0$. As $T \to \infty$, E_{dynamic} converges to $\mu_{AI}^2 + \frac{\alpha\rho\sigma_{AI}^2}{2-\alpha} + \frac{\alpha(1-\rho)\sigma_{AI}^2}{2-\alpha}\sum_i \pi_i^2$.

2. Static AI Noise (Task-Persistent Error): When deliberating on a fixed factual inquiry where querying the foundation model yields a deterministic, persistent answer, $\hat{\boldsymbol{\theta}}_M(t) \equiv \hat{\boldsymbol{\theta}}_M$ ($\eta_{\text{shared}}(t) \equiv \eta_{\text{shared}}$ and $\boldsymbol{\eta}_{\text{indiv}}(t) \equiv \boldsymbol{\eta}_{\text{indiv}}$) for all t. Unrolling Equation 7 with time-invariant advice yields:

$$E_{\text{static}} = (1-\alpha)^{2T}\text{MSE}_{\text{human}}(\rho_h, T) + E_{\text{AI}}^{\text{stat}}(\alpha, \rho, T) \tag{13}$$

where:

$$E_{\text{AI}}^{\text{stat}}(\alpha, \rho, T) = b_T^2\left(\mu_{AI}^2 + \rho\sigma_{AI}^2\right) + (1-\rho)\sigma_{AI}^2\|\mathbf{v}_T\|_2^2 \tag{14}$$

with $\mathbf{v}_T^\top \equiv \sum_{k=0}^{T-1}\alpha(1-\alpha)^k\mathbf{q}_k^\top$. In the static regime, multi-round querying provides no temporal variance dampening, driving collective error directly toward the monoculture ceiling ($\mu_{AI}^2 + \rho\sigma_{AI}^2 \approx 18.40$ at $\rho = 0.90$) for any $\alpha > 0$ as T grows large.

Topology Invariance and Stationary Surrogation. The common-mode AI error is exactly topology-invariant, while the complete finite-time collective error exhibits only small topology-dependent residual differences under the parameter settings considered. As $k \to \infty$, the finite-time projection vector converges asymptotically to the stationary distribution, implying that $\|\mathbf{q}_k\|_2^2 \to \|\boldsymbol{\pi}\|_2^2 = \sum_i \pi_i^2$. We therefore use the stationary quantity as a numerical approximation when its difference from the finite-time projection is negligible. In our finite-horizon setting ($T = 30$), $\|\mathbf{q}_k\|_2^2$ converges rapidly toward this limit: on Watts-Strogatz ($\lambda_2 \approx 0.984$, spectral relaxation time $\tau_{\text{rel}} = \frac{1}{1-\lambda_2} \approx 62$ rounds), $\|\mathbf{q}_k\|_2^2$ increases smoothly from $\|\mathbf{q}_0\|_2^2 = 1/N = 0.001000$ to $\|\mathbf{q}_{30}\|_2^2 \approx 0.001015$ (within 10^{-6} of $\sum\pi_i^2 = 0.001016$); on Erdős-Rényi ($\lambda_2 \approx$

0.64, $\tau_{\text{rel}} \approx 2.8$) $\|\mathbf{q}_{30}\|_2^2 \approx 0.001018$; and on Barabási-Albert ($\lambda_2 \approx 0.86$, $\tau_{\text{rel}} \approx 7.1$) $\|\mathbf{q}_{30}\|_2^2 \approx 0.002474$ (matching $\sum \pi_i^2 = 0.002475$ within 10^{-6}).

Formally, letting $\delta_k \equiv \left|\|\mathbf{q}_k\|_2^2 - \|\boldsymbol{\pi}\|_2^2\right|$, the discrepancy between the exact finite-time arithmetic consensus error $E_c(T)$ and the stationary expression $E_m(T)$ is bounded by $|\Delta E| \leq \sigma_{\text{human}}^2(1-\rho_h)(1-\alpha)^{2T}\delta_T + \alpha^2(1-\rho)\sigma_{AI}^2 \sum_{k=0}^{T-1}(1-\alpha)^{2k}\delta_k$. On Watts-Strogatz and Erdős-Rényi graphs, $\delta_k \leq 1.81 \times 10^{-5}$ ($\delta_{30} \leq 3.1 \times 10^{-7}$), guaranteeing that $|\Delta E| < 0.00004$ across all $\alpha \in [0.01, 1.0]$. On Barabási-Albert scale-free graphs, $\delta_k \leq 1.48 \times 10^{-3}$, bounding $|\Delta E| < 0.00005$ at $\alpha = 0.10$ and < 0.0024 at $\alpha = 1.0$. Consequently, the difference between the exact finite-time error and the stationary expression is negligible across all practical reliance regimes (e.g., at $\alpha = 0.10, \rho = 0.90$, $E_c^{\text{dyn}} = 4.4251$ on WS, 4.4251 on ER, and 4.4258 on BA; stationary $E_m^{\text{dyn}} = 4.4251$ on WS/ER and 4.4262 on BA). In Table 1, the closed-form theory column reports this finite-time evaluation (4.425). The empirical simulation across $3,000$ trials yields 4.569 ± 0.064, a difference of $+0.144$ ($+2.25\sigma$). With a two-sided significance probability of $P(|Z| > 2) \approx 0.0455$ per cell, standard normal variation naturally predicts an expected count of ≈ 0.82 cells exceeding 2σ by chance across 18 cells; an omnibus goodness-of-fit test across all 18 cells ($\chi^2 = 19.2, \text{df} = 18, p = 0.38$) fails to reject the analytical predictions at conventional significance levels.

2.4 Human Error Correlation and Crossover Boundaries

In empirical human groups, individual judgments are rarely orthogonal; decision-makers are subject to correlated estimation errors, common heuristic shortcuts, and shared information sources (Lorenz et al., 2011; Soll and Larrick, 2009; Becker, Brackbill, and Centola, 2017). Let human initial beliefs share pairwise correlation $\rho_h \in [0, 1]$:

$$x_i(0) = \theta + \sqrt{\rho_h}\,\eta_{h,\text{shared}} + \sqrt{1-\rho_h}\,\eta_{h,i} \tag{15}$$

where $\eta_{h,\text{shared}}, \eta_{h,i} \sim \mathcal{N}(0, \sigma_{\text{human}}^2)$ with $\sigma_{\text{human}} = 15.0$ ($\sigma_{\text{human}}^2 = 225.0$). Crucially, because $\mathbb{E}[\eta_{h,\text{shared}}] = 0$, ρ_h parameterizes correlated zero-mean estimation errors across agents rather than systematic directional bias, which we reserve for the non-zero machine alignment offset μ_{AI}.

The expected squared error of the unassisted human consensus ($\alpha = 0$) at finite horizon T evaluates to:

$$\begin{aligned}\text{MSE}_{\text{human}}(\rho_h, T) &= \sigma_{\text{human}}^2\left[\rho_h + (1-\rho_h)\|\mathbf{q}_T\|_2^2\right] \\ &= A_T + B_T\rho_h\end{aligned} \tag{16}$$

where $A_T \equiv \sigma_{\text{human}}^2\|\mathbf{q}_T\|_2^2 = 225\|\mathbf{q}_T\|_2^2$ and $B_T \equiv \sigma_{\text{human}}^2(1-\|\mathbf{q}_T\|_2^2) = 225(1-\|\mathbf{q}_T\|_2^2)$. On Watts-Strogatz small-world graphs at $T = 30$, $\|\mathbf{q}_{30}\|_2^2 \approx 0.001015$, yielding $A_{30} \approx 0.228$ and $B_{30} \approx 224.772$. This evaluates to baseline unassisted errors of 0.229 ($\rho_h = 0.00$), 4.72 ($\rho_h = 0.02$), 11.47 ($\rho_h = 0.05$), and 22.71 ($\rho_h = 0.10$).

Equating the AI-mediated collective error $E(\alpha, \rho, \rho_h, T)$ in Equations 11–13 with the unassisted baseline $\text{MSE}_{\text{human}}(\rho_h, T)$ yields the exact finite-time crossover correlation threshold $\rho_h^*(\alpha)$ above which adopting algorithmic reliance α achieves strictly lower collective error than the unassisted human crowd:

$$\rho_h^*(\alpha) = \frac{E_{\text{AI}}(\alpha, \rho, T) - A_T\left(1-(1-\alpha)^{2T}\right)}{B_T\left(1-(1-\alpha)^{2T}\right)} \tag{17}$$

where $E_{\text{AI}}(\alpha, \rho, T)$ denotes $E_{\text{AI}}^{\text{dyn}}$ or $E_{\text{AI}}^{\text{stat}}$, respectively. Evaluating Equation 17 under monopoly monoculture ($\rho = 0.90, T = 30$) on Watts-Strogatz networks reveals:

- Dynamic Noise Model: At per-step reliance $\alpha = 0.10$ ($w_{\text{cum}} = 95.8\%$), $\rho_h^* \approx 1.87\%$. At moderate per-step reliance $\alpha = 0.40$, $\rho_h^* \approx 3.28\%$. At complete reliance ($\alpha = 1.00$), $\rho_h^* = \frac{18.40 - 0.228}{224.772} \approx 8.08\%$.
- Static Noise Model: At per-step reliance $\alpha = 0.10$ ($w_{\text{cum}} = 95.8\%$), $\rho_h^* \approx 7.42\%$. At moderate per-step reliance $\alpha = 0.40$, rapid saturation to the ceiling ($(1-0.4)^{30} \approx 10^{-7}$) yields $\rho_h^* \approx 8.08\%$. At complete reliance ($\alpha = 1.00$), $\rho_h^* \approx 8.08\%$.

This threshold clarifies a central tension in algorithmic monoculture: the dramatic collective degradation observed under pure monoculture ($\alpha = 1.0$) arises when benchmarking against an idealized, orthogonal human crowd ($\rho_h = 0$). Differentiating Equations 11 and 13 at $\alpha = 0$ yields $\frac{\partial E}{\partial \alpha}\big|_{\alpha=0} = -2T\,\text{MSE}_{\text{human}}(\rho_h) < 0$ for all $\rho_h \geq 0$, establishing that the marginal addition of AI initially decreases collective error even in orthogonal crowds (where a shallow dip exists: dynamic $\alpha^* \approx 0.0017$, error 0.218 vs. 0.229; static $\alpha^* \approx 0.0004$, a -1.2% reduction). When human crowds exhibit correlated heuristics ($\rho_h > 0$), this negative initial gradient is amplified by orders of magnitude (e.g., $-2(30)(22.71) \approx -1362.6$ at $\rho_h = 0.10$), dramatically deepening and widening the U-shaped reliance profile. Combined with the boundary condition $E(1) = 18.40 \gg \min_{\alpha \in [0,1]} E(\alpha)$ (e.g., 18.40 vs. optimal 3.82 at $\rho_h = 0.10$), this negative gradient guarantees that the social optimum α^* lies strictly in the interior $(0, 1)$. Under dynamic noise at per-step $\alpha = 0.10$, AI improves collective accuracy whenever $\rho_h > 1.87\%$, whereas under static task persistence, the AI repeats deterministic errors and improves collective accuracy at per-step $\alpha = 0.10$ only when human correlation exceeds 7.42%. Nevertheless, if reliance is calibrated to the static social optimum (per-step $\alpha^* \approx 0.008$, cumulative reliance $w_{\text{cum}} \approx 20\%$), AI anchors the crowd even at modest correlated human error ($\rho_h = 0.02$), reducing collective error by 20%.

2.5 Unassisted Consensus Across Social Topologies

At $\alpha = 0$, beliefs evolve via pure DeGroot consensus $\mathbf{x}(t) = \mathbf{W}^t\mathbf{x}(0)$, converging asymptotically to $m(0) = \boldsymbol{\pi}^\top\mathbf{x}(0)$ where $\pi_i = \frac{d_i}{\sum_k d_k}$ for the lazy random-walk trust matrix $W_{ii} = 0.5, W_{ij} = 0.5/d_i$. At finite horizon T, the exact unweighted consensus variance is $\sigma^2_{\text{human}}\|\frac{1}{N}\mathbf{1}^\top\mathbf{W}^T\|_2^2$, which expands from the minimal uniform variance $\sigma^2_{\text{human}}/N = 0.225$ at $T = 0$ toward the asymptotic stationary bound $\sigma^2_{\text{human}}\sum_{i=1}^N \pi_i^2$ (0.229 for WS/ER, 0.536 for BA) as T exceeds the spectral relaxation time $\tau_{\text{rel}} = \frac{1}{1-\lambda_2(\mathbf{W})}$.

1. Watts-Strogatz Small-World Graphs: Regular ring lattice with random rewiring ($k = 6, p = 0.1$). Realized degrees yield $\sum \pi_i^2 \approx 0.001016$, predicting an asymptotic baseline error of $225 \times 0.001016 \approx 0.229$ (empirical at $T = 30$: 0.233 ± 0.010). Because high spatial clustering ($C \approx 0.44$; analytic $C(p) = C(0)(1-p)^3 \approx 0.437$) and moderately longer characteristic path lengths (6.1 vs. 2.0 on ER) yield a second eigenvalue of $\lambda_2(\mathbf{W}) \approx 0.984$ (spectral relaxation time $\tau_{\text{rel}} = \frac{1}{1-\lambda_2} \approx 62$ rounds), spectral mixing remains incomplete at finite horizon $T = 30$, explaining why Watts-Strogatz graphs retain non-zero local diversity (1.454 ± 0.011).

2. Erdős-Rényi Random Graphs: Uniform connection probability ($p = 0.05$, $\bar{d} \approx 50$). Realized degrees yield $\sum \pi_i^2 = \frac{1}{N}(1 + \text{Var}(d)/\bar{d}^2) \approx 0.001019$, predicting $225 \times 0.001019 \approx 0.229$ (empirical: 0.231 ± 0.010). Rapid spectral expansion drives diversity to 0.000 well before $T = 30$.

3. Barabási-Albert Scale-Free Graphs: Preferential attachment ($m = 3$). Heavy-tailed degree hubs concentrate stationary weight ($\sum \pi_i^2 \approx 0.00238$ across realized graphs), predicting an elevated baseline error of $225 \times 0.00238 \approx 0.536$, matching the simulated 0.517 ± 0.023 and illustrating the finite-sample influence of prominent agents (Golub and Jackson, 2010).

The pooled theoretical baseline across the three topologies is $(0.229 + 0.229 + 0.536)/3 \approx 0.331$, matching the simulated pooled baseline of 0.327 ± 0.009 within 0.5 standard errors.

3 Model and Simulation Methodology

3.1 Simulation Framework

We simulate networked deliberation across $N = 1,000$ agents structured on graphs $G = (V, E)$ generated via NetworkX (Hagberg, Schult, and Swart, 2008). Parameter sweeps are evaluated over $T = 30$ timesteps across 199,000 total Monte Carlo runs to numerically validate our analytical derivations:[1]

- Experiment 1 (Core Reliance Sweep): 11 reliance levels × 3 correlation regimes ($\rho \in \{0.00, 0.50, 0.90\}$) × 3 topologies (Watts-Strogatz, Erdős-Rényi, Barabási-Albert) × 1,000 independent trials = 99,000 runs ($3,000$ pooled runs per (α, ρ) cell for $\alpha > 0$; baseline $\alpha = 0.0$ reports $1,000$ trials per topology and $3,000$ pooled cross-topology runs).

- Experiment 2 (Human Correlation Sweep): Evaluated on Watts-Strogatz small-world networks across 20 fine reliance levels × 4 human correlation levels ($\rho_h \in \{0.00, 0.02, 0.05, 0.10\}$) under monopoly monoculture ($\rho = 0.90$) × 1,000 independent trials = 80,000 runs.

- Experiment 3 (Static vs. Dynamic Persistence Sweep): Evaluated on Watts-Strogatz small-world networks across 20 fine reliance levels under static machine noise ($\rho = 0.90, \rho_h = 0.00$) × 1,000 independent trials = 20,000 runs (directly compared against the 20,000 dynamic-noise $\rho_h = 0.00$ runs from Experiment 2, yielding 40,000 runs across both arms).

Agent Prior Initialization

At $t = 0$, human agents are initialized with prior beliefs centered around true state $\theta = 100.0$:

$$x_i(0) = \theta + \sqrt{\rho_h}\,\eta_{h,\text{shared}} + \sqrt{1-\rho_h}\,\eta_{h,i} \tag{18}$$

with $\eta_{h,\text{shared}}, \eta_{h,i} \sim \mathcal{N}(0, \sigma^2_{\text{human}})$ and $\sigma_{\text{human}} = 15.0$ ($\sigma^2_{\text{human}} = 225.0$).

Algorithmic Advice Parameterization

We emphasize that chosen parameter values ($\mu_{AI} = 2.0, \sigma_{AI} = 4.0, \sigma_{\text{human}} = 15.0, \rho = 0.90, T = 30, N = 1,000$) serve as stylized order-of-magnitude benchmarks to illustrate qualitative mechanics rather than empirical constants; specific numerical thresholds and optima are parametric functions of this setup, and per-step reliance α^* is defined relative to the $T = 30$ horizon. In our experiments, we evaluate three correlation regimes in Equation 3 ($\mu_{AI} = 2.0$, $\sigma_{AI} = 4.0$, total single-query MSE $= \mu_{AI}^2 + \sigma_{AI}^2 = 20.0$):

- $\rho = 0.90$: Monopoly Monoculture (single centralized model queried across the network).

- $\rho = 0.50$: Correlated Multi-Model (competing frontier models sharing pre-training corpora and alignment pipelines).

[1] Replication code, simulation manifests, and analysis scripts are available at https://github.com/Ashuraj-Herode22/algorithmic-monoculture-sim.

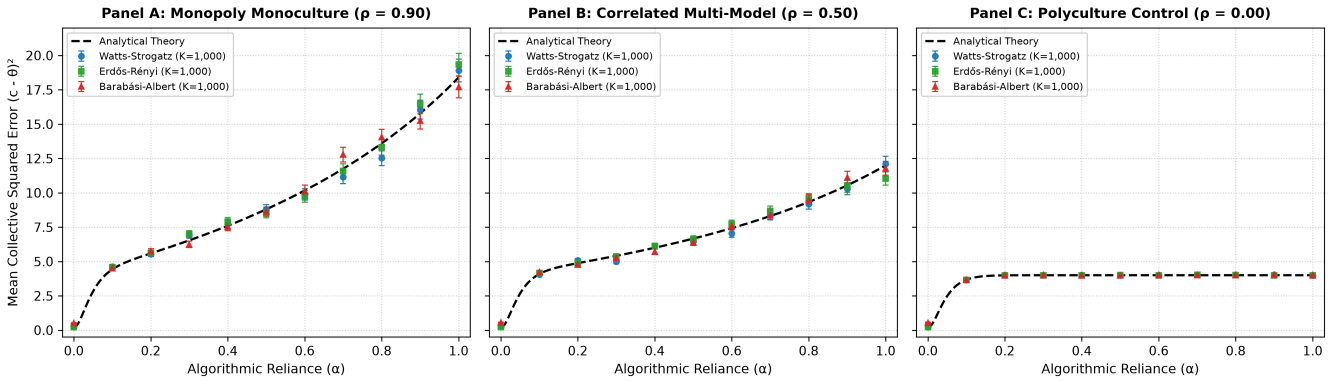


Figure 1: Collective Error vs. Algorithmic Reliance ($N = 1,000, T = 30, K = 1,000$ Trials per Condition, 99,000 Runs). Collective Squared Error $(c(T) - \theta)^2$ as a function of reliance α across independent trials for Watts-Strogatz (blue circles), Erdős-Rényi (green squares), and Barabási-Albert (red triangles) networks at $\rho_h = 0$. Dashed curves overlay the closed-form analytical prediction (Equation 11). Left (Panel A): Polyculture ($\rho = 0.00$), stabilizing at the directional bias floor ($\mu_{AI}^2 = 4.00$). Center (Panel B): Correlated Multi-Model ($\rho = 0.50$), escalating to 11.644 ± 0.293. Right (Panel C): Monopoly Monoculture ($\rho = 0.90$), escalating to 18.648 ± 0.468.

- $\rho = 0.00$: Polyculture Control (independent, variance-decorrelated models sharing only baseline alignment offset μ_{AI}).

DeGroot Trust Matrix

The row-stochastic trust matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ is constructed from graph adjacency matrix $\mathbf{A}$ by assigning self-weight $W_{ii} = 0.5$ and uniform neighbor weights $W_{ij} = 0.5/d_i$ for $j \in \mathcal{N}(i)$.

4 Experimental Results

4.1 Collective Error Escalation and Network Invariance

Figure 1 and Table 1 present the empirical simulation results across 99,000 Monte Carlo runs alongside closed-form analytical predictions.

At $\alpha = 0.0$ with orthogonal priors ($\rho_h = 0$), unassisted human deliberation achieves high collective accuracy (0.327 ± 0.009) through idiosyncratic error cancellation. Disaggregating baseline performance by topology confirms our spectral derivation: Barabási-Albert scale-free networks exhibit higher consensus error (0.517 ± 0.023, predicted 0.536) than Watts-Strogatz (0.233 ± 0.010, predicted 0.229) and Erdős-Rényi (0.231 ± 0.010, predicted 0.229).

Under Monopoly Monoculture ($\rho = 0.90$), collective error escalates from baseline to 18.648 ± 0.468 at $\alpha = 1.0$. This represents a 36- to 81-fold increase depending on network topology, with an average 57-fold escalation relative to the pooled baseline (though crowd error of 18.6 remains 12× lower than unassisted individual error of 225.0 due to baseline scaling as $\mathcal{O}(1/N)$).

Crucially, this error surge is not caused by post-deliberation "diversity loss" per se: unassisted human consensus on connected Erdős-Rényi and Barabási-Albert graphs also converges to zero belief variance (Diversity$(T) = 0.000$) as agents agree, yet achieves minimal error because initial human errors cancel out during aggregation. Rather, collective error escalates under monoculture because external AI advice continuously injects non-canceling directional bias (μ_{AI}) and shared systemic error ($\sqrt{\rho}\eta_{\text{shared}}$) that shift the consensus trajectory $m(t)$ itself.

Across all reliance levels $\alpha \geq 0.1$, the three network topologies collapse onto the identical analytical error trajectory (Figure 1), empirically confirming our proof that row-stochastic networks are transparent to uniform external signals. With 3,000 pooled trials per cell, empirical averages match Equation 11 across the full parameter space: at $\alpha = 0.1$, empirical error (4.569 ± 0.064) tracks theory within 2.3 standard errors (4.42); at $\alpha = 0.6$, empirical error (9.915 ± 0.248) matches theory (10.17); and at $\alpha = 1.0$, empirical error (18.648 ± 0.468) aligns with the theoretical ceiling (18.40).

4.2 Correlated Multi-Model Competition

A common intuition in technological policy is that market competition among multiple foundation model providers will naturally prevent algorithmic monoculture. Our results (Figure 1B, Table 1) demonstrate that when competing models share overlapping web pre-training corpora and alignment protocols such that between-model error correlation remains $\rho = 0.50$, collective error still escalates to 11.644 ± 0.293 at $\alpha = 1.0$ (analytical prediction: 12.00). Moreover, this formulation represents an idealized exchangeable ensemble (effectively an infinite pool of competing models with mu-

Table 1: High-Sample Simulation Results ($N = 1,000$ Agents, $T = 30$ Timesteps, $K = 1,000$ Independent Trials per Condition, 99,000 Total Runs at $\rho_h = 0$). Values denote Mean ± Standard Error of the Mean (SEM) across pooled runs ($3,000$ trials per reliance level across WS, ER, and BA topologies) alongside finite-time closed-form theory (E_c^{dyn}, Equation 11). Baseline $\alpha = 0.0$ rows present $K = 1,000$ trials for each individual topology and $3,000$ pooled cross-topology baseline runs.

Reliance	Polyculture ($\rho = 0.00$)		Multi-Model ($\rho = 0.50$)		Monoculture ($\rho = 0.90$)		Analytical Theory ($T = 30$)		
α	Coll. Error	Diversity	Coll. Error	Diversity	Coll. Error	Diversity	Theory ($\rho = 0$)	Theory ($\rho = 0.5$)	Theory ($\rho = 0.9$)
0.0 (WS Small-World)	Collective Error: 0.233 ± 0.010			Diversity: 1.454 ± 0.011			0.229	0.229	0.229
0.0 (ER Random)	Collective Error: 0.231 ± 0.010			Diversity: 0.000 ± 0.000			0.229	0.229	0.229
0.0 (BA Scale-Free)	Collective Error: 0.517 ± 0.023			Diversity: 0.000 ± 0.000			0.536	0.536	0.536
0.0 (Pooled Baseline)	Collective Error: 0.327 ± 0.009			Diversity: 0.485 ± 0.013			0.331	0.331	0.331
0.1	3.666 ± 0.003	0.224 ± 0.000	4.147 ± 0.048	0.112 ± 0.000	4.569 ± 0.064	0.023 ± 0.000	3.669	4.09	4.425
0.2	3.992 ± 0.003	0.808 ± 0.001	4.908 ± 0.071	0.403 ± 0.000	5.636 ± 0.102	0.081 ± 0.000	3.992	4.88	5.59
0.4	3.997 ± 0.005	2.872 ± 0.002	5.990 ± 0.113	1.436 ± 0.001	7.714 ± 0.171	0.287 ± 0.000	4.004	6.00	7.60
0.6	4.005 ± 0.006	6.037 ± 0.005	7.450 ± 0.163	3.020 ± 0.002	9.915 ± 0.248	0.603 ± 0.000	4.007	7.43	10.17
0.8	4.016 ± 0.008	10.347 ± 0.009	9.420 ± 0.218	5.174 ± 0.004	13.298 ± 0.323	1.035 ± 0.001	4.011	9.33	13.60
1.0 (Pure AI)	$\mathbf{3.997 \pm 0.009}$	$\mathbf{15.980 \pm 0.013}$	$\mathbf{11.644 \pm 0.293}$	$\mathbf{7.995 \pm 0.007}$	$\mathbf{18.648 \pm 0.468}$	$\mathbf{1.599 \pm 0.001}$	**4.016**	**12.00**	**18.40**

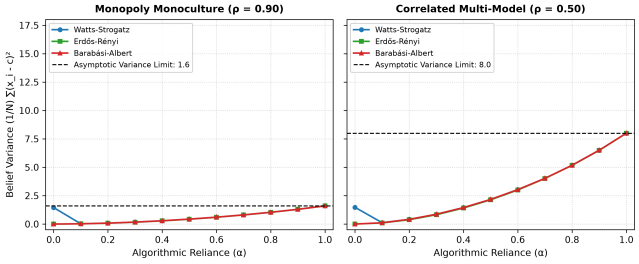


Figure 2: Belief Variance and Residual Dispersion under Algorithmic Mediation ($N = 1,000, T = 30$). Empirical crowd diversity $\frac{1}{N}\sum_i (x_i(T) - c(T))^2$ as a function of reliance α across topologies (Equation 10). Residual belief diversity scales directly with idiosyncratic variance $(1-\rho)\sigma_{AI}^2$ as $\alpha \to 1$, bounding terminal diversity at 1.60 under monopoly monoculture ($\rho = 0.90$) and 8.00 under multi-model competition ($\rho = 0.50$).

tual covariance $\rho = 0.50$). In a realistic oligopolistic market where agents partition their queries evenly across a finite number M of providers with within-provider correlation $\rho_{\text{within}} = 0.90$ (as in monopoly monoculture) and between-provider correlation $\rho_{\text{between}} = 0.50$, the effective pairwise correlation is $\rho_{\text{eff}} = 0.50 + \frac{0.40}{M}$. At complete reliance ($\alpha = 1.0$), collective error evaluates to $\mu_{AI}^2 + \sigma_{AI}^2(0.50 + \frac{0.40}{M}) = 4 + 16(0.50 + \frac{0.40}{M})$, yielding 15.20 for a duopoly ($M = 2$), 14.13 for $M = 3$, and 13.28 for $M = 5$ (interpolating smoothly between 18.40 at $M = 1$ and 12.00 as $M \to \infty$). Market multiplicity with correlated models thus provides limited epistemic protection.

4.3 The Polyculture Control: Independent Advice

Under the polyculture control ($\rho = 0.00$, Figure 1A), idiosyncratic machine variance cancels across the crowd and collective error stabilizes at the squared directional bias floor ($\mu_{AI}^2 = 4.00$), reaching 3.997 ± 0.009 at $\alpha =$ 1.0. While variance decorrelation eliminates the $\rho\sigma_{AI}^2$ penalty, any shared directional alignment offset $\mu_{AI} \neq 0$ persists unattenuated in the crowd consensus.

Table 2: Analytic Evaluation of Optimal Social Reliance (α^*), Error Reduction vs. Unassisted Humans, and Stylized Epistemic Reliance Overshoot Ratio ($\mathcal{R}_{\text{overshoot}}$) Under Dynamic vs. Static AI Noise ($\rho = 0.90, T = 30$). Per-step optimal reliance α^* accumulates to $w_{\text{cum}} \in [51\%, 82\%]$ under dynamic noise and $[20\%, 55\%]$ under static noise over $T = 30$ rounds. Primary $\mathcal{R}_{\text{overshoot}}$ evaluates collective error when agents adopt their regime-specific isolated optimum ($\alpha_{\text{isolated}}^{\text{dyn}} = 0.106$ under dynamic noise; $\alpha_{\text{step}} = 0.080$ under static persistence) relative to the social optimum α^*; parenthetical values report complete per-step reliance ($\alpha_{\text{step}} = 1.0$, MSE = 18.40).

Human Corr. (ρ_h)	Unassisted MSE	Dynamic Noise (Opt α^* / Cut / Ratio)	Static Noise (Opt α^* / Cut / Ratio)
$\rho_h = 0.02$	4.72	0.024 / 51% / 2.0 (8.0)	0.008 / 20% / 4.1 (4.9)
$\rho_h = 0.05$	11.47	0.041 / 72% / 1.4 (5.7)	0.016 / 38% / 2.2 (2.6)
$\rho_h = 0.10$	22.71	0.056 / 83% / 1.2 (4.8)	0.026 / 55% / 1.5 (1.8)

4.4 Dynamic vs. Static AI Noise: Quantitative Regimes

As illustrated in Figure 4, the temporal persistence of machine errors governs the quantitative shape of the reliance trade-off. Table 2 compares the optimal social reliance α^*, percentage error reduction relative to unassisted deliberation, and stylized Epistemic Reliance Overshoot Ratio ($\mathcal{R}_{\text{overshoot}}$) across realistic human correlation levels ($\rho_h \in \{0.02, 0.05, 0.10\}$).

Under dynamic noise, temporal averaging across rounds dampens shared sampling variance by a factor of $\frac{\alpha}{2-\alpha}$, yielding a socially optimal per-step reliance of $\alpha^* \in [0.024, 0.056]$ (cumulative reliance $w_{\text{cum}} \in [51\%, 82\%]$ over $T = 30$ rounds) and reducing collective error by 51%–83%. Under static task persistence, where the model outputs fixed errors across queries, temporal variance dampening disappears; the social optimum shifts

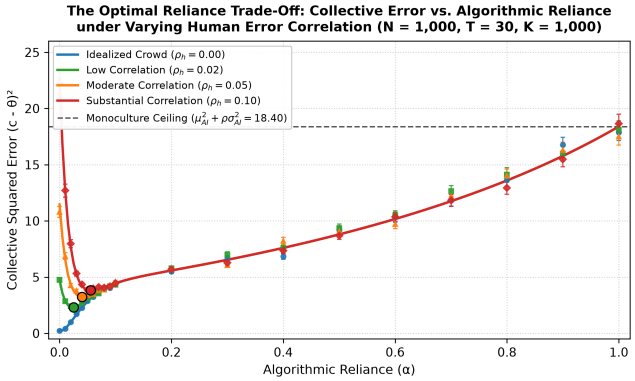


Figure 3: The Optimal Reliance Trade-Off ($N = 1,000, T = 30, K = 1,000$). Collective Squared Error as a function of per-step reliance α across four human correlation regimes ($\rho_h \in \{0.00, 0.02, 0.05, 0.10\}$) on Watts-Strogatz graphs under dynamic AI noise ($\rho = 0.90$). Solid curves denote analytical theory (Equation 11); markers denote empirical Monte Carlo trials ($\pm$ SEM). When human crowd error is correlated ($\rho_h > 0$), collective error exhibits a U-shaped profile: calibrated per-step reliance ($\alpha^* \in [0.024, 0.056]$, cumulative reliance $w_{\text{cum}} \in [51\%, 82\%]$) significantly lowers collective error by anchoring correlated human error, whereas excessive reliance ($\alpha \to 1.0$) drives error up toward the monoculture ceiling (18.40).

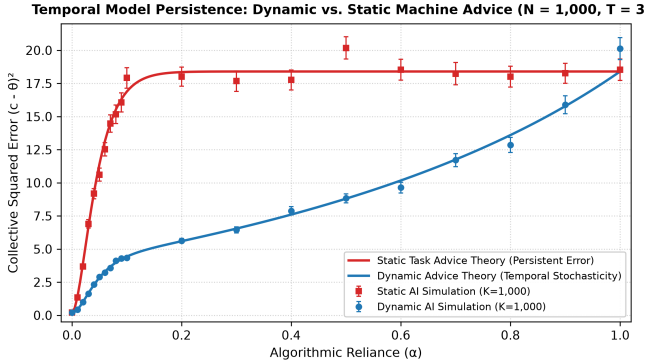


Figure 4: Temporal Model Persistence: Dynamic vs. Static Machine Advice ($N = 1,000, T = 30, K = 1,000$). Comparing collective error under dynamic temporal stochasticity (blue circles, Equation 11) versus persistent task-level error (red squares, Equation 13) on Watts-Strogatz graphs ($\rho = 0.90, \rho_h = 0$). When foundation models repeat deterministic mistakes on a given task, multi-round deliberation provides no temporal variance dampening, rapidly saturating at the monoculture ceiling (18.40).

downward to per-step $\alpha^* \in [0.008, 0.026]$ (cumulative reliance $w_{\text{cum}} \in [20\%, 55\%]$), achieving 20%–55% error reductions.

4.5 Private Benchmarks vs. Epistemic Overshoot

To understand why decentralized agents over-rely on foundation models, consider the decision problem of an isolated individual choosing reliance without social interaction:

- Static Task-Persistent Environment: When deliberating on a fixed factual query where AI advice is deterministic, the agent chooses a single-shot convex combination of their private prior and machine advice:

$$\min_{\alpha} \mathbb{E}[(x_i - \theta)^2] = (1-\alpha)^2\sigma_h^2 + \alpha^2(\mu_{AI}^2 + \sigma_{AI}^2) = 225(1-\alpha)^2 + 20\alpha^2 \tag{19}$$

Differentiating yields the single-shot private optimum:

$$\frac{d}{d\alpha}\mathbb{E}[(x_i - \theta)^2] = 0 \implies \alpha^*_{\text{isolated}} = \frac{225}{245} \approx 0.918 \tag{20}$$

dropping individual error from 225.0 to 18.37 (a 12.25× local accuracy gain). Over $T = 30$ rounds, retaining $(1 - 0.918) = 0.082$ weight on the human prior corresponds to a per-step reliance of $\alpha_{\text{step}} = 1 - (1 - 0.918)^{1/30} \approx 0.080$.

- Dynamic Temporal Stochasticity: When an isolated agent queries the model at each round with independent stochastic draws, multi-round querying averages out idiosyncratic sampling variance. The agent minimizes:

$$\min_{\alpha} \Big[(1-\alpha)^{2T}\sigma^2_{\text{human}} + \mu^2_{AI}(1-(1-\alpha)^T)^2 + \sigma^2_{AI}\frac{\alpha(1-(1-\alpha)^{2T})}{2-\alpha}\Big] \tag{21}$$

For $T = 30$, numerical optimization yields $\alpha^{\text{dyn}}_{\text{isolated}} \approx 0.106$. Here, temporal filtering drives isolated individual error down to 4.89—a 46.0× local accuracy improvement over unassisted heuristics (225.0).

Macro-Level Overshoot Under Hypothesized Correlation Neglect: When decentralized agents enter a connected social network, under the hypothesis that individuals exhibit correlation neglect (cf. DeMarzo, Vayanos, and Zwiebel, 2003; Enke and Zimmermann, 2019)—treating neighbor opinions as independent signals and failing to observe how shared algorithmic reliance propagates through the network—they adopt their private reliance benchmarks ($\alpha_{\text{step}} \approx 0.080$ in static tasks,

$\alpha^{\text{dyn}}_{\text{isolated}} \approx 0.106$ in dynamic tasks) rather than internalizing the positive epistemic externality of unassisted human priors.

Defining the stylized Epistemic Reliance Overshoot Ratio ($\mathcal{R}_{\text{overshoot}}$) as the ratio of collective error under the regime-specific isolated benchmark to minimum collective error at the social optimum α^*:

$$\mathcal{R}_{\text{overshoot}} = \frac{E(\alpha^{\text{bench}}_{\text{isolated}}, \rho, \rho_h, T)}{\min_\alpha E(\alpha, \rho, \rho_h, T)} \tag{22}$$

Unlike the classical game-theoretic Price of Anarchy (Koutsoupias and Papadimitriou, 1999; Roughgarden and Tardos, 2002), which measures the degradation of social welfare in a strategic Nash equilibrium, this ratio captures the epistemic welfare penalty incurred when boundedly rational agents adopt individually optimal reliance benchmarks under correlation neglect (DeMarzo, Vayanos, and Zwiebel, 2003; Enke and Zimmermann, 2019). Recently, Kleinberg, Sinanaj, and Tardos (2026) proved a tight upper bound of 2.0 on the Price of Anarchy in a non-cooperative matching market with algorithmic monoculture. In our continuous networked deliberation setting, decentralized agents do not strategically game the network; rather, under behavioral correlation neglect, they fail to internalize how private reliance choices propagate through social aggregation, adopting their isolated decision-theoretic benchmarks ($\alpha_{\text{step}} \approx 0.080$ in static tasks, $\alpha^{\text{dyn}}_{\text{isolated}} \approx 0.106$ in dynamic tasks). Our stylized ratio thus quantifies the macro-level welfare loss from uncoordinated adoption of locally optimal advice, spanning $[1.5, 4.1]$ under static task persistence and $[1.2, 2.0]$ under dynamic noise (Table 2). At complete per-step reliance ($\alpha_{\text{step}} = 1.0$, MSE = 18.40; parenthetical values in Table 2), the overshoot ratio reaches $[4.8, 8.0]$ relative to the dynamic social optimum and $[1.8, 4.9]$ relative to the static social optimum due to the absence of temporal variance averaging at complete reliance (where $\frac{\alpha}{2-\alpha} = 1$).

Crucially, when human error correlation exceeds $\rho^*_h = 8.08\%$, unassisted human consensus error exceeds the pure monoculture ceiling ($\text{MSE}_{\text{human}} > 18.40$). In this regime, complete reliance ($\alpha_{\text{step}} = 1.0$, MSE = 18.40) does not degrade accuracy below the unassisted human baseline (22.71 at $\rho_h = 0.10$), but instead represents an epistemic overshoot: complete per-step reliance forfeits about 64 percentage points of collective error reduction (77% of achievable gains) relative to the dynamic social optimum at $\rho_h = 0.10$, rising to about 80 percentage points (nearly 100% of achievable gains) just above the 8.08% crossover threshold. Meanwhile, decentralized adoption of the dynamic isolated benchmark ($\alpha^{\text{dyn}}_{\text{isolated}} = 0.106$, error 4.56, a 79.9% reduction) forfeits 3.3 percentage points relative to the social optimum (83.2%, Table 2).

5 Discussion and Sociotechnical Governance

5.1 Individual Auditing vs. Collective Integrity

Current evaluation frameworks for foundation models focus almost exclusively on isolated single-user benchmarks. Our analytical and empirical results demonstrate that micro-level validation is insufficient to guarantee macro-level epistemic safety. A foundation model that reduces individual error by over 12-fold in static tasks (18.37 vs. 225.0) or 46-fold under dynamic sampling (4.89 vs. 225.0) can simultaneously degrade collective accuracy by 57-fold in orthogonal crowds (though absolute crowd error of 18.6 remains 12× lower than unassisted individual error of 225.0 due to baseline error scaling as $\mathcal{O}(1/N)$) or induce an epistemic overshoot of up to 4.1× at the private reliance benchmark and up to 8.0× at complete per-step reliance ($\alpha_{\text{step}} = 1.0$) in correlated crowds.

5.2 Algorithmic Diversity as Public Infrastructure

Preserving distributed collective intelligence requires treating statistical independence across AI systems as public epistemic infrastructure:

1. True Epistemic Pluralism: Training ensembles of models on curated, non-overlapping data partitions and distinct alignment objectives to drive inter-model error correlation $\rho \to 0$ and center directional offsets ($\mathbb{E}[\mu_{AI}] = 0$). We note that while distinct data pipelines and alignment architectures reduce correlation, statistical independence cannot be assumed a priori and must be empirically validated across tasks.

2. Calibrated Reliance Protocols: Institutional workflows (e.g., clinical diagnostics, intelligence forecasting, peer review) should require participants to record independent human priors before consulting foundation models, bounding per-step reliance—illustratively near $\alpha^* \in [0.01, 0.06]$ under our stylized benchmark parameters (cumulative reliance $w_{\text{cum}} \in [26\%, 84\%]$ over $T = 30$ rounds)—to prevent unassisted signals from being prematurely drowned out.

3. Epistemic Stubbornness vs. Passive Unassisted Nodes: Because row-stochastic networks satisfy $(\mathbf{I} - (\mathbf{I} - \mathbf{D}_\alpha)\mathbf{W})\mathbf{1} = \mathbf{D}_\alpha\mathbf{1}$ on any connected graph with non-zero AI adoption ($\mathbf{D}_\alpha = \text{diag}(\boldsymbol{\alpha}) \neq \mathbf{0}$), merely setting $\alpha_i = 0$ for a subset of nodes only delays the characteristic timescale of bias absorption (τ_{abs}). Unassisted nodes passively average their neighbors, ultimately absorbing 100% of external machine bias μ_{AI} as t grows. Crucially, merely designating nodes as socially stubborn ($W_{ii} = 1$) is insufficient if those agents continue to consult AI

($\alpha_i > 0$), because their belief updates $x_i(t+1) = (1-\alpha_i)x_i(t) + \alpha_i\hat{\theta}_{M,i}(t)$ continue to incorporate external machine advice, inexorably pulling them toward the AI estimate. True long-term structural resilience requires epistemically stubborn human anchors that combine zero AI reliance ($\alpha_i = 0$) with persistent belief anchoring—either through complete social insulation ($W_{ii} = 1$ with $\alpha_i = 0$) or Friedkin-Johnsen persistence (Friedkin and Johnsen, 1990; Acemoglu et al., 2013) where $x_i(t+1) = (1-\alpha_i)[(1-\gamma_i)\sum_j W_{ij}x_j(t) + \gamma_i x_i(0)] + \alpha_i\hat{\theta}_{M,i}(t)$, which at $\alpha_i = 0$ reduces to permanent attachment to their initial pre-AI human prior ($x_i(t+1) = (1-\gamma_i)\sum_j W_{ij}x_j(t) + \gamma_i x_i(0)$ with $\gamma_i > 0$). Such anchors permanently pin the stationary opinion profile away from pure machine monoculture.

6 Limitations and Future Work

First, our model parameterizes foundation model errors via Gaussian structural residuals with scalar correlation $\rho \in [0,1]$; future empirical work should estimate full error covariance matrices from multi-LLM benchmark residuals (Kim et al., 2025). Second, while we analyze homogeneous reliance α to establish exact mean-field bounds, real-world networks exhibit heterogeneous, degree-correlated adoption (α_i scaling with node centrality d_i), where hub adoption disproportionately accelerates monocultural lock-in. Third, our framework models machine errors $\boldsymbol{\eta}_{AI}$ and human cognitive heuristics $\boldsymbol{\eta}_h$ as mutually independent random vectors ($\mathbb{E}[\boldsymbol{\eta}_{h,\text{shared}}\eta_{\text{shared}}] = 0$); in practice, because foundation models are pre-trained on corpora of human text that encode common human cognitive biases, machine and human errors may be positively cross-correlated, which would further diminish the epistemic benefits of AI advice. Finally, extending DeGroot linear updating to Bayesian agents who dynamically estimate model correlation represents a promising avenue for theoretical extension.

7 Conclusion

This study formalizes the Algorithmic Monoculture Paradox: while foundation models provide substantial accuracy gains for isolated decision-makers, uncoordinated reliance in connected networks accumulates shared machine bias and erodes the wisdom of crowds. When human decision-makers suffer from correlated human errors ($\rho_h > 0$), collective error follows a U-shaped trajectory where calibrated per-step AI reliance ($\alpha^* \in [0.008, 0.056]$, corresponding to cumulative reliance weights of $w_{\text{cum}} \in [20\%, 82\%]$ over $T = 30$ rounds) significantly improves collective judgment, whereas individually rational reliance overshoots the social optimum.

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